

2.7.3. Stochastic Differential Equations

We consider ODE $\frac{dx}{dt} = f(t, x)$

$\xi = \xi(t)$: disturbance with stochastic behavior

$$\rightarrow \frac{dx}{dt} = F(t, x, \xi)$$

We look at special form $\frac{dx}{dt} = \overbrace{f(t, x)}^{\text{deterministic drift coeff}} + \underbrace{g(t, x)}_{\text{noise intensity}} \underbrace{N(t)}_{\text{Gaussian white noise process}}$

$$\rightarrow x(t) = x(t_0) + \int_{t_0}^t f(s, x(s)) ds + \int_{t_0}^t g(s, x(s)) N(s) ds$$

⊗

not defined

$$\int_{t_0}^t g(s, x) dW(s), \quad W: \text{Wiener process}$$

Ito stochastic integral

$$\int_a^b g(s, \omega) dW(s, \omega), \quad g: \text{stoch. proc.}, \quad W: \text{Wiener proc.}$$

$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$: complete prob. space.

① $t_1 < t_2 \Rightarrow \mathcal{F}_{t_1} \subset \mathcal{F}_{t_2}$

② $W(t, \omega) : \mathcal{F}_t\text{-meas.}, \quad \omega \in \mathbb{R}_+$

③ $t \leq t_i < t_{i+1} \Rightarrow W(t_{i+1}, \omega) - W(t_i, \omega) : \text{indep of } \mathcal{F}_t$

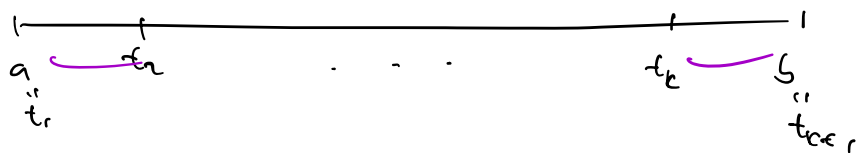
$$L^2[a, b] = \{ \text{real valued rand proc } g(t) \text{ defined on } [a, b] \}$$

$$: \quad g(t, \omega) : \mathcal{F}_t\text{-meas.}, \quad \int_a^b \int g(t, \omega)^2 dt < \infty \quad \text{w.p. 1} \}$$

for a simple function $g(t, \omega)$, say $a = t_1 < t_2 < \dots < t_{n+1} = b$: partition,

$$g(t, \omega) = g(t_i, \omega) \text{ for } t \in [t_i, t_{i+1}) \quad : \quad \mathcal{F}_{t_i} \text{-meas}, \text{ w.d.}$$

$$\rightarrow \int_a^b g(t, \omega) dW(t) := \sum_{i=r}^k g(t_i, \omega) [W(t_{i+1}) - W(t_i)]$$



$\rightarrow$ simple process g_n .

mesh size $\rightarrow 0$

$$\Rightarrow \int_a^b g(t, \omega) dW(t) := \lim_{n \rightarrow \infty} \int_a^b g_n(t, \omega) dW(t) \quad \text{in } L^2$$

$$\left(\mathbb{E} \int_a^b (g(t, \omega) - g_n(t, \omega))^2 dW(t) \rightarrow 0 \right) \quad (?)$$

$$\text{If } g \in L_{ad}([a, b]; \mathbb{R}^d) \text{ and } \int_a^b \mathbb{E} \|g(t)\|^2 dt < \infty,$$

$$\text{then } \begin{cases} \mathbb{E} \left[\int_a^b g(t, \omega) dW(t) \right] = 0 \\ \mathbb{E} \left[\left\| \int_a^b g(t, \omega) dW(t) \right\|^2 \right] = \int_a^b \mathbb{E} \|g(t)\|^2 dt \end{cases}$$

$$(*) \quad X(t) = X(t_0) + \int_{t_0}^t f(s, X(s)) ds + \int_{t_0}^t g(s, X(s)) dW(s)$$

$$\Leftrightarrow dX(t) = f(t, X(t)) dt + g(t, X(t)) dW(t) \quad (\text{SDE of It\^o type})$$

$$\text{Def } \underbrace{f(t, \omega) \in L_{ad}(\Omega; L^1[a, b])}_{\text{a stoch. proc.}} \quad \text{if } \begin{cases} \mathcal{F}_t \text{-adapted} \\ \int_a^b \|f(t, \omega)\|^p dt < \infty \quad \text{a.s.} \end{cases}$$

$$\text{Def } \mathbb{R}^d \text{-valued stoch. proc. } X(t) = X(t; t_0, x_0) : \text{ solution of}$$

$$\begin{cases} dx(t) = f(t, x(t))dt + g(t, x(t))dW(t) & , t \in [t_0, T] \quad \text{--- (i)} \\ x(t_0) = x_0 & \text{--- (ii)} \end{cases}$$

where $W(t) = (W_1(t), \dots, W_n(t))^T \in \mathbb{R}^m$, $x_0: F_{t_0}$ -meas, $E[\|x_0\|^2] < \infty$

if ① $x(t): \text{cont.}, F_t$ -adapted

② $f \in \mathcal{L}_{ad}(\Omega; L^1[a, b])$ $\mathbb{R}^n$ -valued,

$g \in \mathcal{L}_{ad}(\Omega; L^2[a, b])$ $\mathbb{R}^{n \times m}$ -valued

③ $x(t)$ satisfies (i) w.p.1.

④ x satisfies (ii) w.p.1.

a sol x is unique if another sol y is indistinguishable $(P(x(t)=y(t), \forall t \in [t_0, T]) = 1)$

Def (Ito formula) $x(t) = x(t; t_0, x_0)$ $\mathbb{R}^n$ -dim F_t -adapted stoch proc w/

$$x(t) = x(t_0) + \int_{t_0}^t f(s, x(s))ds + \int_{t_0}^t \sigma(s, x(s))dW(s) \quad \text{a.s.}$$

$$V \in C^{1,2}(\mathbb{R}_t \times \mathbb{R}^n; \mathbb{R})$$

$\Rightarrow V(t, x(t))$: stochastic process w/

$$V(t, x(t)) = V(t_0, x(t_0)) + \int_{t_0}^t \mathcal{L}V(s, x(s))ds + \int_{t_0}^t V_x(s, x(s))\sigma(s, x(s))dW(s) \quad \text{a.s.}$$

where $\mathcal{L}V(t, x(t)) = V_t(t, x(t)) + V_x(t, x(t))f(t, x(t)) +$

$$\frac{1}{2} \text{tr} [\sigma^T(t, x(t)) V_{xx}(t, x(t)) \sigma(t, x(t))]]$$

In differential forms, $dx(t) = f(t, x(t)) dt + \sigma(t, x(t)) dW(t)$ a.s., $V \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R})$

$$\Rightarrow dV(t, x(t)) = \mathcal{L}V(t, x(t)) dt + V_x(t, x(t)) \sigma(t, x(t)) dW(t) \text{ a.s.}$$

the operator $\mathcal{L}$ is called the averaged derivative
(infinitesimal diffusion generator)

$$\mathcal{L}V(t, x) = \lim_{h \rightarrow 0} \frac{1}{h} \left[\mathbb{E}[V(t+h, x(t+h))] - V(t, x) \right]$$

stochastic systems with time delay

$$\begin{cases} dx(t) = f(t, x_t) dt + g(t, x_t) dW(t), & t \in [t_0, T] \\ x_{t_0}(s) = \phi(s), & s \in [-r, 0] \end{cases}$$

$\hookrightarrow$ given as stoch. proc.

$$\|\phi\|_r = \sup_{s \in [-r, 0]} \|\phi(s)\|$$

$$\|x_t\|_r = \sup_{s \in [-r, t]} \|x(s)\|$$

$\phi: F_{t_0}$ -meas., conti random variable $[-r, 0] \rightarrow \mathbb{R}^n$, $\mathbb{E}[\|\phi\|_r^p] < \infty$
for some $p > 0$

Def The trivial sol $x \equiv 0$ is said to be

① almost-surely stable if $\forall \varepsilon, \varepsilon' > 0, t_0 \in \mathbb{R}_+, \exists \delta = \delta(\varepsilon, \varepsilon', t_0)$:

$$\|\phi\|_r < \delta \Rightarrow \mathbb{P}\{\omega: \sup_{t \geq t_0} \|x(t)\| > \varepsilon'\} < \varepsilon$$

② pth moment stable if $\forall \varepsilon > 0, t_0 \in \mathbb{R}_+, \exists \delta = \delta(\varepsilon, t_0)$:

$$(\|\phi\|_r^p < \delta) \Rightarrow \mathbb{E} \left[\sup_{t \geq t_0} (\|x(t)\|)^p \right] < \varepsilon$$

③ almost-surely asymptotically stable if ① and

$$\mathbb{P} \{ \omega : \lim_{t \rightarrow \infty} \sup_{s \geq t} (\|x(s)\|) = 0 \} = 1$$

↪ ?

④ pth moment asymptotically stable if ② +

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\sup_{s \geq t} (\|x(s)\|)^p \right] = 0$$

⑤ pth moment exponentially stable if $\exists p, K, \lambda > 0$? $\forall t_0 \in \mathbb{R}_+$,

$$(\|\phi\|_r^p < \delta \Rightarrow \mathbb{E} [\|x(t)\|^p] \leq K \|\phi\|_r^p e^{-\lambda(t-t_0)})$$

"globally" if $\delta = \infty$

"uniformly" if δ indep of t_0

2.7.4. Comparison Methods for Stochastic Systems

SDE of Ito type

$$\begin{cases} dx(t) = f(t, x(t))dt + g(t, x(t))dW(t) \\ x(t_0) = x_0 \end{cases}$$

$$f \in C(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R}^n), \quad g \in C(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R}^{n \times m}), \quad W: \text{ Wiener process}$$

Thm (Comparison theorem) $V \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R})$

$$D^+V(t, x) \leq h(t, V(t, x)) \quad \text{a.s. for all } (t, x) \in \mathbb{R}_+ \times \mathbb{R}^n,$$

$h \in C(\mathbb{R}_+ \times \mathbb{R}^+; \mathbb{R})$, $h(t, z)$: concave and quasi-monotone decreasing in z for all $t \in \mathbb{R}_+$. ?

$$r(t) = r(t; t_0, u_0) : \text{maximal sol of } \begin{cases} \dot{u}(t) = h(t, u) \\ u(t_0) = u_0 \end{cases}$$

$$\mathbb{E}[V(t_0, x_0)] \leq u_0 \Rightarrow \mathbb{E}[V(t, x(t))] \leq r(t; t_0, u_0)$$

Then (Stability theorem) $a \in K_c, b \in K_v$

$$V \in C^{1,2}(\mathbb{R}_+ \times S(e); \mathbb{R}), \quad S(e) \subset \mathbb{R}^n$$

$$\forall (t, x) \in \mathbb{R}_+ \times S(e), \quad \begin{cases} b(t, \|x\|) \leq V(t, x) \leq a(t, \|x\|) \quad \text{a.s.} \\ \mathcal{L}V(x, t) \leq h(t, V(t, x)) \quad \text{a.s.} \end{cases}$$

$p \geq 1$, $h \in C(\mathbb{R}_+ \times \mathbb{R}_+; \mathbb{R})$, $h(t, 0) \equiv 0$, $h(t, z)$: concave, quasi-monotone
 increasing in z

$\Rightarrow$ stability properties of $u \equiv 0$ imply the corresponding
 p th moment stability of $x \equiv 0$

Stochastic impulsive systems

$$\begin{cases} dx(t) = f(t, x(t))dt + g(t, x(t))dw(t) & , \quad t \neq \tau_k \\ \Delta x(t) = I(t, x(t)) & , \quad t = \tau_k \\ x(t_0) = x_0 \end{cases}$$

2.8. Stoch Impul System with Time Delay (SISD)

Notation & Assumption

$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$: complete

$\{\mathcal{F}_t\}_{t \geq 0}$: filtration, right-continuous, $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets

$\omega(t) = (\omega_1(t), \dots, \omega_m(t))^T$: m-dim Wiener process

$\mathcal{C}([r, 0]; \mathbb{R}^n), \mathcal{PC}([r, 0]; \mathbb{R}^n)$

$x: [t-r, \infty) \rightarrow \mathbb{R}^n \quad x_t: [r, 0] \rightarrow \mathbb{R}^n \quad x_t(s) = x(t+s)$

$x_{t-} \in \mathcal{PC}([r, 0]; \mathbb{R}^n) \quad \begin{cases} x_{t-}(s) = x(t+s) & -r \leq s < 0 \\ x_{t-}(0) = x(t^-) \end{cases}$

$p > 0, \mathcal{L}_{\mathcal{F}_0}^p([r, 0]; \mathbb{R}^n)$: the set of all $\mathcal{F}_0$ -meas'

$\mathcal{PC}([r, 0]; \mathbb{R}^n)$ -valued random variables

$\phi = \{\phi(s) \mid -r \leq s \leq 0\} \quad \omega \mid \mathbb{E}[\|\phi\|_r^p] \leq c$

Def $a, b \in \mathbb{R}, a < b, D \subset \mathbb{R}^n$

$\psi: [a, b] \times \Omega \rightarrow D \in \mathcal{PC}([a, b] \times \Omega; D)$ (or D -cadlag)

if for almost all $\omega \in \Omega$,

$$\begin{cases} \psi(t^+, \omega) = \psi(t, \omega) & \forall t \in [a, b) \\ \psi(t^-, \omega) \text{ exists in } D, & \forall t \in (a, b] \\ \psi(t^-, \omega) = \psi(t, \omega) & \text{for all but finitely many pts } t \in [a, b] \end{cases}$$

$$\psi: [a, \infty) \times \Omega \rightarrow D \in \mathcal{PC}([a, \infty) \times \Omega; D)$$

$$\text{if } \forall c > a, \quad \psi(t, \omega) \in \mathcal{PC}([a, c] \times \Omega; D) \text{ where } t \in [a, c]$$

$$\begin{cases} dx(t) = f(t, x_t) dt + g(t, x_t) dW(t) & \text{SDE w/ time delay} \\ x_{t_0} = \phi(s), \quad s \in [v, 0] \end{cases} \text{ where}$$

$$\phi \in \mathcal{L}_{F_0}^2([v, 0]; \mathbb{R}^n)$$

$$\Rightarrow x(t) = \phi(0) + \int_{t_0}^t f(s, x_s) ds + \int_{t_0}^t g(s, x_s) dW(s)$$

$$\mathbb{E} \left[\int_{t_0}^t g(s, x_s) dW(s) \right] = 0, \quad \mathbb{E} \left(\left\| \int_{t_0}^t g(s, x_s) dW(s) \right\|^2 \right) = \int_{t_0}^t \mathbb{E} \|g(s, x_s)\|^2 ds$$

SISD

$$\begin{cases} dx(t) = f(t, x_t) dt + g(t, x_t) dW(t), & t \neq \tau_k(x_{t-}) \\ \Delta x(t) = I(t, x_{t-}) & t = \tau_k(x_{t-}) \end{cases} \quad \text{--- (x)}$$

$\tau_k \in \mathcal{C}^2(\mathbb{R}^n; \mathbb{R}_+)$: impulsive hyper-surface

$$0 = \tau_0(x) < \tau_1(x) < \dots, \quad \lim_{k \rightarrow \infty} \tau_k(x) = \infty$$

We assume the sd : right-contin & $\Delta x = x(t) - x(t-)$: $\mathcal{F}_{t-}$ -adapted

Def $\mathbb{R}^n$ -valued rand proc $x \in \mathcal{PC}([t_0, t_0 + \alpha]; \mathbb{R}^n)$: sd of (x) if

$$\textcircled{1} \quad \mathcal{T} = \{t \in [t_0, t_0 + \alpha] \mid t = \tau_k(x_{t-}) \text{ for some } k\} : \text{finite}$$

② $x(t)$: cont. for $t \in (t_0, t_0 + \alpha]$ ($\mathbb{T}$), F_t -adapted

③ $f \in L_{ad}(\Omega; L[t_0, t_0 + \alpha])$, $g \in L_{ad}(\Omega; L^2[t_0, t_0 + \alpha])$

④ $t \in (t_0, t_0 + \alpha]$, $\phi \in \mathcal{L}_{F_0}^2([t_0, t_0 + \alpha]; \mathbb{R}^n)$, $I(t_k, x_{t_k^-})$: F_{t_k} -adapted

$$x(t) = \begin{cases} \phi(t - t_0) & t \in [t_0 - \tau, t_0] \\ \phi(t) + \int_{t_0}^t f(s, x_s) ds + \int_{t_0}^t g(s, x_s) dW(s) + \sum_{\{k: t_k \in (t_0, t]\}} I(t_k, x_{t_k^-}) & t \in (t_0, t_0 + \alpha] \end{cases}$$

w.p. 1

⑤ $\forall t \in \mathbb{T}$, $x(t)$ satisfies (*) w.p. 1.

⑥ $x(t)$ satisfies the initial condition w.p. 1.

Special case : impulsive instants occur at fixed times $t = t_k$.

2.9. Switched Systems

$$2.9.1. \begin{cases} \dot{x} = f(t, x) + u(t) \\ x(t_0) = x_0 \end{cases}, \quad u(t) = \sum_{k=0}^{\infty} C_k x(t) \mathbb{1}_{[t_k, t_{k+1})}(t).$$

the controller $u(t)$ switches values at $t = t_k$ (switching controller)

$$\Rightarrow \begin{cases} \dot{x} = f(t, x) + C_k x, & t \in [t_k, t_{k+1}) \\ x(t_0) = x_0 \end{cases} \quad \text{switched system.}$$

general nonlinear switched system

$$\begin{cases} \dot{x} = f_{\sigma(t)}(t, x) & , t \geq t_0 \\ x(t_0) = x_0 \end{cases}$$

$\sigma: [t_0, \infty) \rightarrow [N]$, N : # of subsystems

↳ piecewise constant function, switching signal / law / rule

Q ① Sufficient conditions to guarantee asymp stab of a switched system for arbitrary switching signal

✓ ② Identify the switching signals for which a switched system is asymp stable.

✓ ③ Construct a switching signal that makes the system stable.

Stability of single mode $\longrightarrow$ Stability of switched mode

① Enough dwell time $\checkmark$ $I_{inf}(\tau) = \{ \tau \mid \inf(t_k - t_{k-1}) \geq \tau, \forall k \}$

② Sufficiently large average dwell time τ_{avg}

$$I_{avg}(\tau, N_0) : \forall T \geq t \geq t_0, N_\sigma(t, T) \leq N_0 + \frac{T-t}{\tau_{avg}}$$

δ
of switching moments of σ in (t, T)

③ (General than ①) Minimum switching

σ : right-continuous Markov chain (process),

takes values with generator $P = (p_{ij})_{i,j \in N}$

$$P\{\sigma(t+h)=j \mid \sigma(t)=i\} = \begin{cases} p_{ij}h + o(h) & i \neq j \\ 1 + p_{ii}h + o(h) & i = j \end{cases}$$

$h > 0$, $p_{ij} > 0$: transition rate from i to j , $i \neq j$

$$p_{ii} = - \sum_{j \neq i} p_{ij}$$

→ switched system
$$\begin{cases} \dot{x}(t) = f(t, x(t), \sigma(t)) \\ x(t_0) = x_0, \sigma(t_0) = \sigma_0 \end{cases}$$

2.9.2.
$$\begin{cases} \dot{x} = A_i x & t \in [t_{k-1}, t_k) \\ x(t_0) = x_0 \end{cases} \quad t_{k-1} < t_k \nearrow \infty$$

Thm A_i : Hurwitz, $i \in I$

$x \equiv 0$: exponentially stable if $\lambda_{\min} - \nu(t_k - t_{k-1}) \leq 0$, $\forall k \in \mathbb{N}$,

where $\mu = \frac{\lambda_M}{\lambda_m}$ w/ $\lambda_M = \max\{\lambda_{\max}(P_i), i \in I\}$, $\lambda_m = \min\{\lambda_{\min}(P_i), i \in I\}$

P_i : positive definite, $A_i^T P_i + P_i A_i = -Q_i$ for any pos-def Q_i ,

$$0 < \nu < \alpha_i = \frac{c_i}{\lambda_M} \quad c_i > 0: \quad \frac{\partial V_i}{\partial x} A_i x \leq -c_i \|x\|^2$$

(F) $V_i(x) = x^T P_i x \Rightarrow \lambda_m \|x\|^2 \leq V_i(x) \leq \lambda_M \|x\|^2$ — (A)

$$\dot{V}_i(x) = \frac{\partial V_i}{\partial x} A_i x \leq -c_i \|x\|^2 \leq -\frac{c_i}{\lambda_M} V_i(x) = -\lambda_i V_i(x)$$

$$\Rightarrow V_i(x(t)) \leq e^{-\lambda_i(t-t_{k-1})} V_i(x(t_{k-1}))$$

$$\textcircled{d}) \Rightarrow \forall i, j, \quad V_j(x(t)) \leq \mu V_i(x(t)), \quad \mu = \frac{\lambda_M}{\lambda_m}$$

$\textcircled{e}) [t_0, t_1) \rightarrow \text{Subsystem 1}, [t_1, t_2) \rightarrow \text{Subsystem 2}$

$$\left(\begin{aligned} V_1(x(t)) &\leq e^{-\lambda_1(t-t_0)} V_1(x_0) \\ V_2(x(t)) &\leq e^{-\lambda_2(t-t_1)} V_2(x(t_1)) \\ &\leq e^{-\lambda_2(t-t_1)} \mu V_1(x(t_1)) \\ &\leq \mu e^{-\lambda_2(t-t_1)} e^{-\lambda_1(t_1-t_0)} V_1(x_0) \end{aligned} \right)$$

$$t \in [t_{k-1}, t_k) \Rightarrow V_i(x(t)) \leq \mu^{\overset{k-1}{i-1}} e^{-\lambda_i(t-t_{k-1})} e^{-\lambda_{i-1}(t_{k-1}-t_{k-2})} \dots e^{-\lambda_1(t_1-t_0)} V_1(x_0)$$

$$\text{Let } \lambda := \min \{ \lambda_i : i \in I \}$$

$$\Rightarrow V_i(x(t)) \leq \mu^{i-1} e^{-\lambda(t-t_0)} V_1(x_0)$$

$$= \mu^{i-1} e^{-\nu(t-t_0)} e^{-(\lambda-\nu)(t-t_0)} V_1(x_0)$$

$$\textcircled{?} \leq \mu^{i-1} e^{-\nu(t-t_0)} \quad "$$

...

$$\leq \mu^{i-1} e^{-\nu(t-t_0)} \dots e^{-\nu(t_k-t_{k-1})} V_1(x_0) e^{-(\lambda-\nu)(t-t_0)}$$

$$= \underbrace{\mu e^{-\nu(t_1-t_0)} \dots \mu e^{-\nu(t_k-t_{k-1})} V_1(x_0)}_{\leq 1} "$$

$$\Rightarrow V_i(x(t)) \leq V_1(x_0) e^{-(\lambda-\nu)(t-t_0)} \quad t \geq t_0$$

$$\Rightarrow \|x(t)\| \leq \underbrace{\left(\frac{k}{\sqrt{\mu}} \right)}_{\text{or } \sqrt{\mu}} \|x_0\| e^{-\frac{(\lambda-\nu)(t-t_0)}{2}} \quad t \geq t_0$$

□

$$* t_k - t_{k-1} > \frac{\log \mu}{\nu} = \textcircled{c} \text{ dwell time}$$

* average dwell time $\approx$ feedback!

2.10. Stochastic Switched Systems w/ time delay

$$\begin{cases} dx(t) = f_{\sigma(t)}(t, x_t) dt + g_{\sigma(t)}(t, x_t) dW(t), & t \geq t_0 \\ x_{t_0}(s) = \phi(s), & s \in [-\nu, 0] \end{cases} \quad (\Phi)$$

Def $\mathbb{R}^n$ -valued random process $x(t) = x(t; t_0, \phi)$

$(x(t), \sigma(t))$: solution of SSD (Φ) if

① $x(t)$: conti, adapted w.r.t. $\{F_t\}_{t \geq t_0}$

② $f_{\sigma(t)}(t, x_t) \in \mathcal{L}_{ad}(\Omega; L[t_0, T])$, $g_{\sigma(t)}(t, x_t) \in \mathcal{L}_{ad}(\Omega; L^2[t_0, T])$,

$$x(t) = \phi + \int_{t_0}^t f_{\sigma(s)}(s, x_s) ds + \int_{t_0}^t g_{\sigma(s)}(s, x_s) dW(s) \quad \text{w.p.1}$$

$$x(t) = \phi(t), \quad \forall t \in [-\nu, 0]$$

Thm $\sigma: \mathbb{R}_+ \rightarrow I$ switching signal

$$\exists c > 0 : \|f_{\sigma(t)}(t, \psi)\|^2 + \|g_{\sigma(t)}(t, \psi)\|^2 \leq c(1 + \|\psi\|_{r^2}) \quad \text{a.s.}$$

$$\forall t \in \mathbb{R}_+, \psi \in C([-\nu, 0]; \mathbb{R}^n),$$

$$\|f_{\sigma(t)}(t, \psi_1) - f_{\sigma(t)}(t, \psi_2)\|^2 + \|g_{\sigma(t)}(t, \psi_1) - g_{\sigma(t)}(t, \psi_2)\|^2 \leq c\|\psi_1 - \psi_2\|_{r^2}^2$$

$$\text{a.s. } \forall t \in \mathbb{R}_+, \psi_1, \psi_2 \in C([-\nu, 0]; \mathbb{R}^n)$$

$$\Rightarrow \left\{ \begin{array}{l} \exists! \text{ solution } x \text{ defined for } t \geq -r \text{ w/ initial fn } \phi \in \mathcal{L}_{F_0}^p([r, 0]; \mathbb{R}^n) \\ \mathbb{E} \left[\sup_{-r \leq t \leq T} \|x(t)\|^2 \right] < \infty, \quad \forall T > 0 \end{array} \right.$$

If σ : Markov process indep of the Wiener process, SSSP

$$\left\{ \begin{array}{l} dx(t) = f(t, x_t, \sigma(t))dt + g(t, x_t, \sigma(t))dW(t) \\ x_0(s) = \phi(s), \quad s \in [r, 0] \\ \sigma(t_0) = \sigma_0 \end{array} \right. \quad \text{--- (22)}$$

Thm $\sigma: \mathbb{R}_+ \rightarrow I$ switching signal, Markov process

$$\exists c > 0 : \|f(t, \varphi, \sigma(t))\|^2 + \|g(t, \varphi, \sigma(t))\|^2 \leq c(1 + \|\varphi\|^2) \text{ a.s.}$$

$$\forall t \in \mathbb{R}_+, \varphi \in C([r, 0]; \mathbb{R}^n)$$

$$\|f(t, \varphi_1, \sigma(t)) - f(t, \varphi_2, \sigma(t))\|^2 + \|g(t, \varphi_1, \sigma(t)) - g(t, \varphi_2, \sigma(t))\|^2$$

$$\leq c\|\varphi_1 - \varphi_2\|^2 \text{ a.s.}$$

$$\forall t \in \mathbb{R}_+, \varphi_1, \varphi_2 \in C([r, 0]; \mathbb{R}^n)$$

$$\Rightarrow \left\{ \begin{array}{l} \exists! \text{ solution } x(t), t \geq -r \text{ w/ initial fn } \phi \in \mathcal{L}_{F_0}^p([r, 0]; \mathbb{R}^n) \\ \mathbb{E} \left[\sup_{-r \leq t \leq T} \|x(t)\|^2 \right] < \infty, \quad \forall T > 0 \end{array} \right.$$

Def (Generalized Itô formula) $x: [-r, \infty) \rightarrow \mathbb{R}^n$ Itô process satisfying (22),

$$V(t, x, i) \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n \times I; \mathbb{R}_+) \text{ w/ } \sigma = i \in I$$

$\Rightarrow V(t, x, i)$: Itô process

$$dV(t, x_t, i) = \mathcal{L}V(t, x_t, i)dt + V_x(t, x_t, i)g(t, x_t, i)dW(t) \text{ a.s.}$$

$$\mathcal{L}V(t, x_t, \bar{i}) = V_t(t, x_t, \bar{i}) + V_{x_t}(t, x_t, \bar{i}) f(t, x_t, \bar{i})$$

$$+ \sum_{j=1}^N \lambda_{ij} [g^T(t, x_t, \bar{i}) V_{x_t}(t, x_t, \bar{i}) g(t, x_t, \bar{i})]$$

$$+ \sum_{j=1}^N \lambda_{ij} V(t, x_t, j) \quad a.s.$$

Def $x \equiv 0$ is

① stable in pth moment if $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon, p) > 0 : \mathbb{E}[\|\phi(t, \cdot)\|^p] < \delta$

② unif stable " if " $\delta = \delta(\varepsilon)$ " $\Rightarrow \mathbb{E}[\|\phi(t, \cdot)\|^p] < \varepsilon$

③ asymp stable " if ① + $\exists \eta, \eta(\varepsilon) > 0 : \mathbb{E}[\|\phi(t, \cdot)\|^p] < \eta$

$$\Rightarrow \lim_{t \rightarrow \infty} \mathbb{E}[\|\phi(t, \cdot)\|^p] = 0$$

④ unif asymp stable " if ② + $\exists \eta > 0 : \forall \varepsilon > 0, \exists T = T(\eta, \varepsilon) > 0 :$

$$\sim \Rightarrow \mathbb{E}[\|\phi(t, \cdot)\|^p] < \varepsilon, \forall t \geq t_0 + T$$

⑤ exp stable " if $\exists K > 0, \lambda > 0 : \mathbb{E}[\|\phi(t, \cdot)\|^p] \leq K \mathbb{E}[\|\phi(t_0, \cdot)\|^p] e^{-\lambda(t-t_0)}$

$$\Leftrightarrow \mathbb{E}[\|\phi(t, \cdot)\|^p] < \eta$$

+ "globally" when $\delta, \eta = \infty$.

impulsive-switched system / switched system w/ impulsive effects

Impulse arise when the system switches from one mode to another.

A nonlinear deterministic impulsive-switched system :

$$\begin{cases} \dot{x}(t) = f(t, x(t)), & t \neq t_k \\ \Delta x(t) = I(t, x(t)), & t = t_k \\ x(t_0) = x_0 \end{cases} \quad (\Phi \Phi \Phi)$$

$\sigma: [t_0, \infty) \rightarrow I$: switching signal, piecewise constant

switching moments $t_1 < t_2 < \dots < \infty$

$$\sigma(t) = i_k \text{ for } t \in [t_{k-1}, t_k) \quad \Rightarrow \quad \dot{x}(t) = f_{i_k}(t, x(t)) \text{ , } t \in [t_{k-1}, t_k)$$

Def $x \in PC([t_0 - \alpha, t_0 + \alpha]; \mathbb{R}^n)$ ($\alpha > 0$), σ : switching signal

$(x, \sigma) \in PC(\sigma)$: sol of (1) if

① $x \in PC$: conti for all $t \in \mathbb{R}_+$ except at switching (impulsive) moments $T = \{t_k\}_{k \geq 1}$.

② $\dot{x}$ exists, x conti for $t \neq t_k$, at $t = t_k$ right-hand derivative exists

③ $\dot{x}$ satisfies (1) for $t \in \mathbb{R}_+ \setminus T$

④ x " (2) for $t \in T$

⑤ x " (3)

general form of solution :
$$x(t) = \Phi(t, t_0) \int_{t_0}^t f_{i_k}(s, x(s)) ds + \sum_{\{k: t_0 < t_k \leq t\}} I(t_k, x(t_k^-))$$

2.1. Singularly Perturbed Systems

large-scale interconnected systems $\rightarrow$ dynamics with different speeds or multiple timescale

$$\begin{cases} \dot{x} = f(t, x, z) \\ \dot{z} = G(t, x, z) \end{cases}$$

$x \in \mathbb{R}^m$: slow variable, $z \in \mathbb{R}^n$: fast variable

In short time, x : nearly constant

In long time, z : reaches quasi-steady state

$$\begin{cases} \dot{x}_s = f(t, x_s, z_s) \\ 0 = G(t, x_s, z_s) \end{cases} \quad x_s, z_s : \text{quasi-steady states}$$

perturbation parameter ε

$$\begin{cases} \dot{x} = f(t, x, z) \\ \varepsilon \dot{z} = g(t, x, z) \end{cases}, \quad g = \varepsilon G \quad \text{with } 0 < \varepsilon \ll 1$$

$$\varepsilon = 0 \rightarrow \begin{cases} \dot{x} = f(t, x, h(t, x)) \\ 0 = g(t, x, z) \end{cases}, \quad h(t, x) : \text{solution of } 0 = g(t, x, z)$$

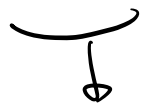
Method : view system as an interconnection of lower order subsystems

→ study the stability of each individual subsystem

→ combine with the connection between subsystems viewed as perturbations

m th order interconnected system $\dot{x}_i = f_i(t, x_i) + g_i(t, x_i)$, $i = 1, \dots, m$

$$x_i \in \mathbb{R}^{n_i}, \quad \sum_{i=1}^m n_i = n, \quad x = (x_1^T, \dots, x_m^T)^T$$


 interconnection

Assume $f_i(t, 0) = 0$, $g_i(t, 0) = 0$

Ignoring the interconnection, $\dot{x}_i^0 = f_i(t, x_i)$

Assume $x_i \equiv 0$: uniformly asymp stable

$$V(t, x) = \sum_{i=1}^m d_i V_i(t, x_i)$$

$$\dot{V}(t, x) = \underbrace{\sum_{i=1}^m d_i \left[\frac{\partial V_i}{\partial t} + \frac{\partial V_i}{\partial x_i} f_i(t, x_i) \right]}_{\text{negative definite}} + \sum_{i=1}^m d_i \frac{\partial V_i}{\partial x_i} g_i(t, x)$$

($\therefore V_i$: Lyapunov for asymp stable subsystem)

Assume $\left\{ \begin{array}{l} \frac{\partial V_i}{\partial t} + \frac{\partial V_i}{\partial x_i} f_i(t, x_i) \leq -\alpha_i \phi_i^2(x_i) \quad , \quad t \geq t_0 \\ \left\| \frac{\partial V_i}{\partial x_i} \right\| \leq \beta_i \phi_i(x_i) \quad , \end{array} \right.$

$\alpha_i, \beta_i > 0$, ϕ_i : positive-definite

Suppose $\|g_i(t, x)\| \leq \sum_{j=1}^m \gamma_{ij} \phi_j(x_j)$, $i=1, \dots, m$, $\gamma_{ij} \geq 0$

$$\Rightarrow \dot{V}(t, x) \leq \sum_{i=1}^m d_i \left[-\alpha_i \phi_i^2(x_i) + \sum_{j=1}^m \beta_i \gamma_{ij} \phi_i(x_i) \phi_j(x_j) \right]$$

$$\dot{V}(t, x) \leq -\frac{1}{2} \phi^T (DS + S^T D) \phi, \quad \phi = (\phi_1, \dots, \phi_m)^T, \quad D = \text{diag}(d_1, \dots, d_m),$$

$$S = (s_{ij}), \quad s_{ij} = \begin{cases} \alpha_i - \beta_i \gamma_{ij} & , \quad i=j \\ -\beta_i \gamma_{ij} & , \quad i \neq j \end{cases}$$

If D chosen such that $DS + S^T D > 0$, the asymp stab guaranteed.

If S : M-matrix, $\exists$ diagonal D .

Def $n \times n$ matrix S : M-matrix if $\det \begin{pmatrix} s_{11} & \dots & s_{1n} \\ \vdots & & \vdots \\ s_{n1} & \dots & s_{nn} \end{pmatrix} > 0$, $\forall k=1, \dots, n$.

Lemma $\exists$ positive diagonal $D : DS + S^T D > 0 \Leftrightarrow S$: Hurwitz.

Thm

2.12. Some inequalities.

Jensen $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ convex, $x: \Omega \rightarrow \mathbb{R}$ rv, $\mathbb{E}[|x|] < \infty$

$$\Rightarrow \varphi(\mathbb{E}[x]) \leq \mathbb{E}[\varphi(x)]$$

Chebyshev $x: \Omega \rightarrow \mathbb{R}^n$ rv, $\mathbb{E}[|x|^p] < \infty$, $p > 0$

$$\Rightarrow \mathbb{P}[\omega : |x| \geq \varepsilon] \leq \frac{\mathbb{E}[|x|^p]}{\varepsilon^p}, \quad \forall \varepsilon > 0$$

Hölder $x, y: \mathbb{R}^n$ -valued rv, $p, q \in (1, \infty)$ $\frac{1}{p} + \frac{1}{q} = 1$

$$|\mathbb{E}[x^T y]| \leq \mathbb{E}[|x|^p]^{\frac{1}{p}} \mathbb{E}[|y|^q]^{\frac{1}{q}}$$

Bihari $t \in [0, T]$, $T > 0$, $u(t) \geq 0$ Bvcl meas', $v(t) \geq 0$ integrable.

$k: \mathbb{R}_+ \rightarrow \mathbb{R}_+$: cont' nondecreasing fcn, $k(t) > 0$, $\forall t > 0$

$$\text{If } \exists c > 0 : |u(t)| \leq c + \int_0^t v(s) k(u(s)) ds \quad \forall t \in [0, T],$$

$$\text{then } |u(t)| \leq G^{-1} \left(G(c) + \int_0^t v(s) ds \right)$$

$$G(c) + \int_0^t v(s) ds \in \text{Dom}(G^{-1}),$$

$$G(r) = \int_0^r \frac{ds}{k(s)}$$

x, y : $\mathbb{R}^d$ -valued rand proc. P_x, P_y

Prokhorov distance btw P_x, P_y : $D(x, y) = D(P_x, P_y)$

$D(x, y) = 0 \rightarrow x, y$ have same prob meas.

$$P\{\omega : \lim_{n \rightarrow \infty} \|x_n(\omega) - x(\omega)\| = 0\} = 1 \Rightarrow \{x_n\} : D\text{-Cauchy seq}$$

Skorokhod's theorem $\{x_n\} : D\text{-Cauchy}$

$$\rightarrow \text{can construct } \{y_n\}, y : \text{r.v.s} \quad \begin{cases} D(x_n, y_n) = 0 \\ P\{\omega : \lim_{n \rightarrow \infty} \|y_n(\omega) - y(\omega)\| = 0\} = 1 \end{cases}$$

Def $Q = \{x_n : n \in \mathbb{N}\}$: totally D -bounded if

$\forall$ infinite seq $\{x_n\} \subset Q$ has a D -Cauchy subseq

Prokhorov's thm Q : totally D -bdd $\Leftrightarrow \forall \varepsilon > 0, \exists$ opt set $K_\varepsilon \subset \mathbb{R}^d$ s.t.

$$P\{x \in K_\varepsilon\} > 1 - \varepsilon, \quad \forall x \in Q$$